

1. The Salary Structure (Gross Salary)

The system first calculates the **Gross Salary** (the total amount an employee earns before any deductions). This calculation is triggered automatically in `Employee.java`.

Component Breakdown:

- **Basic Salary:** The foundational amount set during registration.
- **HRA (House Rent Allowance):** Fixed at **40%** of the Basic Salary.
- **Conveyance & Medical:** These default to **5%** of the Basic Salary each, unless a specific custom value is provided.
- **Special Allowance:** Defaults to **10%** of the Basic Salary if no custom value is entered.

The Formula:

Gross Salary = Basic + HRA + Conveyance + Medical + Special

2. Statutory and Fixed Deductions

Once the Gross Salary is established, the system identifies what needs to be taken out. These are categorized as **Total Deductions**.

- **EPF (Employee Provident Fund):** Calculated as **12%** of the Basic Salary.
- **Professional Tax (PT):** This is a location-specific fee defined in the code:
 - **Gurugram:** 0
 - **Delhi/Pune:** 200
 - **Noida:** 150
 - **Bangalore:** 250
- **Voluntary/Variable Deductions:** Includes **Health Insurance** and **TDS** (Tax Deducted at Source) as entered during registration.

The Formula:

Total Deductions = EPF + Health Insurance + Professional Tax + TDS

3. Attendance-Based Adjustments

The system doesn't just pay the Gross Salary; it adjusts it based on how many days the employee actually worked. This logic lives in `EmployeeService.java`.

1. **Daily Rate:** The system finds the value of a single workday:

$\text{Per Day Salary} = \text{Gross Salary} / \text{Total Days in Month}$

2. **Earned Salary:** The system multiplies the daily rate by the number of days the employee was present:

$\text{Earned Salary} = \text{Per Day Salary} * \text{Present Days}$

4. The Final Payout (Net Pay)

The final step is subtracting the fixed deductions from the attendance-adjusted earned salary. This is the actual amount the employee receives in their bank account.

The Final Formula:

$\text{Final Salary} = \text{Earned Salary} - \text{Total Deductions}$

Numerical Example Walkthrough

If an employee has a **Basic Salary of 50,000** and works **27 out of 30 days** in **Noida**:

1. **Calculate Gross:** * Basic (50k) + HRA (20k) + Conv (2.5k) + Med (2.5k) + Spec (5k)
= **80,000 Gross.**
2. **Calculate Deductions:**
 - EPF (6k) + PT (150) + Insurance/TDS (e.g., 4.5k) = **10,650 Total Deductions.**
3. **Adjust for Attendance:**
 - $(80,000 / 30 \text{ days}) * 27 \text{ days present} = \mathbf{72,000 \text{ Earned Salary.}}$
4. **Final Pay:**
 - $72,000 - 10,650 = \mathbf{61,350 \text{ Final Payout.}}$